

SIMATS ENGINEERING ASSESSMENT TOOLS

COURSE NAME: Theory of Computation COURSE CODE: CSA1304

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 10%

QUESTIONS: 10 Total Marks:100 Data Interpretation

Code of Conduct:

I **Yuvann Nithish R(19241167)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Yuvann Nithish R

SIMATS ENGINEERING

ASSESSMENT TOOLS

Given Data (Grammar)

Consider the CFG:

$S \rightarrow aSb \mid ab$

Q1. Language Identification

Based on the given grammar, which of the following strings belong to the language? A) ab

B) aabb

C) aaabbb

D) abb

Q2. Pattern Recognition

From the grammar, identify the correct pattern:

A) $a^n b^n$

B) $(ab)^n$

C) $a^n b^m$

D) $(a+b)^*$

Q3. Derivation Analysis

Which of the following is a valid derivation for aabb?

A) $S \rightarrow aSb \rightarrow aaSbb \rightarrow aabb$

B) $S \rightarrow ab \rightarrow aabb$

C) $S \rightarrow aSb \rightarrow ab$

D) $S \rightarrow aaSbb \rightarrow aabb$

Q4. Missing Step Identification

Fill in the missing step:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow \underline{\hspace{2cm}} \rightarrow aabb$

A) ab

B) S

C) ϵ

D) aSb

Q5. Parse Tree Interpretation

How many levels are there in the parse tree for string aaabbb?

A) 2

B) 3

C) 4

D) 5

Consider:

$S \rightarrow SS \mid a$

Q6. Ambiguity Check

This grammar is:

A) Unambiguous

B) Ambiguous

C) Regular

D) Deterministic

Q7. String Generation

Which string can be generated?

A) a

B) aa

C) aaa

D) All of the above

Q8. Interpretation of Ambiguity

Ambiguity means:

A) One string $\rightarrow$ one parse tree

B) One string $\rightarrow$ multiple parse trees

C) No derivation

D) Infinite symbols

Given Data (Grammar 1)

$S \rightarrow 0S1 \mid 01$

Q9. Derivation Interpretation

Which of the following derivations correctly generates the string 0011? A) $S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$

B) $S \rightarrow 01 \rightarrow 0011$

C) $S \rightarrow 0S1 \rightarrow 01$

D) $S \rightarrow 00S11 \rightarrow 0011$

Given Data (Grammar 2)

$S \rightarrow aS \mid Sb \mid ab$

Q10. Structural Analysis

Which of the following strings can be generated by the grammar? A) ab

- B) aab
- C) abb
- D) All of the above

SOLUTIONS:

Q1. Language Identification

Grammar: $S \rightarrow aSb \mid ab$

Answer: A, B, C

A) $ab \rightarrow S \rightarrow ab$

B) $aabb \rightarrow S \rightarrow aSb \rightarrow a(ab)b \rightarrow aabb$

C) $aaabbb \rightarrow S \rightarrow aSb \rightarrow aaSbb \rightarrow aa(ab)bb \rightarrow aaabbb$

Q2. Pattern Recognition

Answer: A) $a^n b^n$

The rule $S \rightarrow aSb$ adds **one a and one b** each time.

Examples:

$S \rightarrow ab$

$S \rightarrow aSb \rightarrow aabb$

$S \rightarrow aSb \rightarrow aaSbb \rightarrow aaabbb$

Therefore, the number of as is always equal to the number of bs.

Pattern = $a^n b^n$

Q3. Derivation Analysis

Answer: A

$S \rightarrow aSb$

$\rightarrow aaSbb$

$\rightarrow aa(ab)bb$

$\rightarrow aabb$

Therefore, option A gives the correct derivation.

Q4. Missing Step Identification

Answer:

Given:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow \underline{\hspace{2cm}} \rightarrow aabb$

If we replace S with ab:

$aaSbb \rightarrow aa(ab)bb$

which gives:

$aaabbb$

not $aabb$.

So none of the given options correctly completes this derivation.

The correct derivation for $aabb$ is:

$S \rightarrow aSb \rightarrow a(ab)b \rightarrow aabb$

Q5. Parse Tree Interpretation

Answer: D) 5

For $aaabbb$:

S

↓

aSb

↓

aaSbb

↓

aaaSbbb

↓

aaabbb

There are **5 levels** when the terminal level is included.

Q6. Ambiguity Check

Answer: B) Ambiguous

Grammar:

$S \rightarrow SS \mid a$

The string aaa can have different parse structures.

For example:

(a)(aa)

or

(aa)(a)

Therefore, the same string can have **multiple parse trees**.

So the grammar is **ambiguous**.

Q7. String Generation

Answer: D) All of the above

Grammar:

$S \rightarrow SS \mid a$

a

$S \rightarrow a$

aa

$S \rightarrow SS \rightarrow aa$

aaa

$S \rightarrow SS \rightarrow aSS \rightarrow aaa$

Therefore, **a**, **aa**, and **aaa** can all be generated.

Q8. Interpretation of Ambiguity

Answer: B) One string $\rightarrow$ multiple parse trees

A grammar is called **ambiguous** when at least one string can be represented by **more than one parse tree**.

Therefore:

One string $\rightarrow$ Multiple Parse Trees ✓

Q9. Derivation Interpretation

Answer: A

Grammar:

$S \rightarrow 0S1 \mid 01$

To generate 0011:

$S \rightarrow 0S1$

$\rightarrow 00S11$

Now replace S with 01:

$\rightarrow 00(01)11$

$\rightarrow 0011$

Therefore, **A is correct**.

Q10. Structural Analysis

Answer: D) All of the above

Grammar:

$S \rightarrow aS \mid Sb \mid ab$

ab

$S \rightarrow ab$

aab

$S \rightarrow aS \rightarrow a(ab) \rightarrow aab$

abb

$S \rightarrow Sb \rightarrow (ab)b \rightarrow abb$

SIMATS ENGINEERING ASSESSMENT TOOLS

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand